HARSHAL SURESH RAUT, Ph.D.

Computational Fluid Dynamics | Fluid-Structure Interaction | Multiphase Flow | Scientific Machine Learning
Postdoctoral Researcher, Purdue University | West Lafayette, Indiana, USA | hraut@purdue.edu | Google Scholar | LinkedIn

RESEARCH PROFILE

Computational fluid dynamicist and multiphysics researcher developing high-fidelity numerical methods and parallel solvers for unsteady aerodynamics and hydrodynamics, fluid-structure interaction, multiphase flow, phase change, particle-laden flow, transport in biological tissues, and scientific machine learning. Research record: 11 published peer-reviewed journal articles, one issued patent, and solver development in C/C++, CUDA, MPI, OpenFOAM/CFDEM, FEniCS/FEniCSx, and Python. I use high-fidelity computation to identify physical mechanisms and develop reduced models, scaling laws, and control strategies.

Research interests: computational aerodynamics and hydrodynamics; unsteady fluid-structure interaction and aeroelasticity; wave-assisted and bioinspired propulsion; flow control; multiphase flow and phase change; scientific machine learning; high-performance computing; computational biofluids and transport.

ACADEMIC APPOINTMENTS

Purdue University | West Lafayette, IN 01/26-Present
Postdoctoral Researcher | Advisor: Prof. Hector Gómez
- Develop computational models for particle-laden injectable formulations using coupled CFD-DEM and immersed-boundary methods, including plunger-driven microparticle transport, cohesion, wall interactions, clogging, and injectability.
- Develop finite-element models for pressure-driven and advection-diffusion transport in anisotropic biological tissues using FEniCS/FEniCSx and MRI/DTI-informed material properties.

Johns Hopkins University | Baltimore, MD 12/22-12/25
Postdoctoral Researcher, Flow Physics and Computation Laboratory | Advisor: Prof. Rajat Mittal and Prof. Jung-Hee Seo
- Developed and used high-fidelity immersed-boundary fluid-structure interaction solvers to study wave-induced flapping-foil propulsion, leading-edge vortex physics, multi-foil wake interactions, foil design, and irregular-wave forcing.
- Derived an interpretable leading-edge-vortex-based thrust model from more than 450 simulations and used energy maps and force partitioning to connect flow structures with thrust production and design.
- Developed a 3D VOF-based multiphase-flow capability and integrated it with an immersed-boundary FSI solver; achieved approximately 85% parallel efficiency on 400 CPU cores.

Indian Institute of Technology Bombay | Mumbai, India 01/18-10/22
Ph.D. Research Fellow, Computational Fluid Dynamics Laboratory | Advisor: Prof. Atul Sharma and Prof. Amitabh Bhattacharya
- Developed a sharp-interface dual-grid level-set and immersed-boundary solver for multifluid-structure interaction with phase change, including parallel C++/CUDA/MPI implementations.
- Established synchronization and lock-on mechanisms in actively forced nucleate boiling; proposed strategies for heat-transfer enhancement and sustained boiling in zero gravity.
- Developed a dual-grid deep-reinforcement-learning/CFD framework that reduced training cost by a factor of 5.5 while retaining flow-control accuracy.

EDUCATION

Indian Institute of Technology Bombay | Mumbai, India **2018-2022**
Ph.D., Mechanical Engineering
Thesis: Computational Multi-Fluid Structure Interaction Study on Nucleate Boiling in Presence of Externally Actuated Surfaces | CGPA: 10/10

Indian Institute of Technology Hyderabad | Hyderabad, India **2011-2016**
B.Tech. (Honors) and M.Tech. Dual Degree, Mechanical and Aerospace Engineering
M.Tech. thesis: Flow past in-line oscillating square cylinder: Direct numerical simulations and linear stability theory | M.Tech. thesis grade: 10/10; Overall CGPA: 8.13/10

SELECTED RESEARCH CONTRIBUTIONS

- **Wave-assisted propulsion and unsteady fluid-structure interaction:** Established a physics-to-design framework for passive-pitch flapping-foil propulsors through high-fidelity FSI simulations, energy-transfer maps, force partitioning, a leading-edge-vortex thrust model, multi-foil wake exploitation, pitch-limiting strategies, foil-shape optimization, and irregular-wave performance. The framework provides a foundation for unsteady aerodynamics and aeroelasticity questions involving flexible lifting surfaces and propulsors.
- **Multiphase flow, boiling, and thermal transport:** Developed sharp-interface numerical methods for boiling with moving solids and phase change; identified lock-on between bubble departure and surface motion; studied 3D multi-site boiling; proposed a patented oscillation-based method to sustain nucleate boiling in zero gravity.
- **Scientific machine learning for PDEs and flow control:** Developed a multi-fidelity dual-grid DRL-CFD method for active flow control and heat-transfer enhancement; also taught a guest lecture on physics-informed neural networks (PINNs) for solving PDEs.
- **High-performance multiphysics solver development:** Built and parallelized CFD/FSI/multiphase solvers using MPI, CUDA, immersed-boundary methods, level-set/VOF interface capturing, finite-volume and finite-element methods; current work extends to CFD-DEM and DTI-informed biological transport.

PEER-REVIEWED JOURNAL PUBLICATIONS

Author name and article title shown in bold. Publications listed in reverse chronological order by first online publication date.

1. **Raut, H. S.**; Seo, J.-H.; Mittal, R. **Hydrodynamics of flapping foils undergoing irregular motion with application to wave-assisted propulsion.** *Journal of Fluid Mechanics*, 1041, A12 (2026). DOI
2. **Raut, H. S.**; Seo, J.-H.; Mittal, R. **Harnessing leading-edge vortices for improved thrust performance of wave-induced flapping foil propulsors.** *Physical Review Fluids*, 10(8), 084705 (2025). DOI
3. **Raut, H. S.**; Seo, J.-H.; Mittal, R. **Dynamics and thrust performance of a modeled multifoil wave-induced flapping foil propulsor.** *Ocean Engineering*, 317, 119930 (2025). DOI
4. **Raut, H. S.**; Seo, J.-H.; Mittal, R. **Hydrodynamic performance and scaling laws for a modelled wave-induced flapping-foil propulsor.** *Journal of Fluid Mechanics*, 999, A1 (2024). DOI
5. **Raut, H. S.**; Bhattacharya, A.; Sharma, A. **Numerical study on 3D nucleate boiling over an oscillating base plate with high wall superheat and multiple nucleation sites.** *Numerical Heat Transfer, Part A: Applications*, 87(1), 2372470 (2026; published online 2024). DOI
6. **Raut, H. S.**; Bhattacharya, A.; Sharma, A. **A dual grid-based deep reinforcement learning and computational fluid dynamics method for flow control and its application to nucleate pool boiling.** *International Journal of Heat and Mass Transfer*, 227, 125561 (2024). DOI
7. **Raut, H. S.**; Bhattacharya, A.; Sharma, A. **Computational multifluid-structure interaction study on nucleate boiling under the effect of stationary or oscillating torus.** *International Journal of Heat and Mass Transfer*, 193, 122995 (2022). DOI
8. **Raut, H. S.**; Bhattacharya, A.; Sharma, A. **Sustaining nucleate boiling in zero gravity using asymmetric sinusoidal base-plate oscillation.** *International Journal of Heat and Mass Transfer*, 184, 122262 (2022). DOI
9. Nguyen, T.-B.; Liu, D.; **Raut, H. S.**; Bhattacharya, A.; Sharma, A.; Tran, T. **Enhancement of convective heat transfer using magnetically flapping fin array.** *International Communications in Heat and Mass Transfer*, 129, 105638 (2021). DOI
10. **Raut, H. S.**; Bhattacharya, A.; Sharma, A. **Dual grid level set method based direct numerical simulations of nucleate boiling with oscillating base plate.** *International Journal of Thermal Sciences*, 162, 106785 (2021). DOI
11. **Raut, H. S.**; Dixit, H. N. **Vortex shedding patterns in flow past a streamwise oscillating square cylinder at low Reynolds number using dynamic meshing.** *Physics of Fluids*, 31, 113605 (2019). DOI

PATENT

1. **Raut, H. S.**; Bhattacharya, A.; Sharma, A. **Method and boiling system for providing nucleate boiling of fluid at zero gravity by surface oscillations.** Indian Patent No. 507595, issued 2024.

RESEARCH EXPERIENCE

Purdue University | West Lafayette, IN 01/26-Present
Postdoctoral Researcher | Advisor: Prof. Hector Gómez
- CFD-DEM for injectable particle suspensions: coupled Eulerian fluid flow with Lagrangian particles; particle-particle and particle-wall contact/cohesion; moving plunger represented by a penalization-based immersed-boundary method; emphasis on clogging, segregation, and injectability.
- Computational transport in brain tissue: Darcy-type pressure-driven flow and advection-diffusion transport in anisotropic porous tissue; DTI-based tensor mapping, registration, FEniCS/FEniCSx implementation, and coupled 3D-1D transport concepts.

Johns Hopkins University | Baltimore, MD 12/22-12/25
Postdoctoral Researcher | Advisor: Prof. Rajat Mittal and Prof. Jung-Hee Seo
- Studied wave-assisted propulsion (WAP), where free-to-pitch submerged hydrofoils convert wave-induced heave into thrust.
- Used sharp-interface immersed-boundary FSI simulations, energy maps, and force partitioning to identify leading-edge vortices as the dominant thrust-producing mechanism and derive a phenomenological thrust model.
- Extended the framework to multi-foil systems, pitch constraints, foil geometry, and irregular waves; showed how wake interactions and nonperiodic kinematics can improve thrust.
- Developed a 3D VOF capability and integrated it with an immersed-boundary FSI solver on structured nonuniform grids; implemented MPI parallelization and high-throughput post-processing for multi-terabyte datasets.

Indian Institute of Technology Bombay | Mumbai, India 01/18-10/22
Ph.D. Research Fellow | Advisor: Prof. Atul Sharma and Prof. Amitabh Bhattacharya
- Developed an in-house sharp-interface dual-grid level-set solver for incompressible two-phase flow with phase change and moving immersed solids.
- Studied oscillating surfaces and immersed structures as active mechanisms for controlling bubble departure, synchronization, and boiling heat transfer; extended the method to 3D multi-site boiling.
- Proposed asymmetric surface oscillations for sustaining nucleate boiling under zero gravity; work resulted in an issued patent.
- Developed a dual-grid DRL-CFD framework for active flow control and nucleate boiling, reducing computational training cost by approximately 5.5x relative to the fine-grid-only strategy.

Indian Institute of Technology Hyderabad | Hyderabad, India 10/17-12/17
Junior Research Fellow, Fluid Physics Laboratory | Advisor: Prof. Harish N. Dixit
- Modeled the thin air film beneath Euler's disk using lubrication equations and implemented a reduced-order finite-difference solver in C++.

University of British Columbia | Vancouver, Canada 09/16-05/17
Research Assistant, Department of Mathematics | Advisor: Prof. Neil Balmforth and Prof. Mark Martinez
- Developed reduced-order mathematical models for dewatering of pulp suspensions in piston-cylinder and screw-press systems using Darcy-type transport equations; implemented solvers in MATLAB.

Indian Institute of Technology Hyderabad | Hyderabad, India 08/15-04/16
Graduate Student Researcher, Fluid Physics Laboratory | Advisor: Prof. Harish N. Dixit
- Performed direct numerical simulations of flow past a streamwise oscillating square cylinder using dynamic meshing in OpenFOAM and analyzed wake instability using Galerkin approximation, Floquet theory, and numerical integration.

TEACHING EXPERIENCE AND PEDAGOGY

Guest Lecture 11/25
Physics-Informed Neural Networks for Solving Partial Differential Equations
- Delivered a lecture for a numerical methods course connecting PDE formulation, neural-network approximation, automatic differentiation, residual-based loss functions, boundary conditions, and verification against classical numerical solutions.

Indian Institute of Technology Bombay | Mumbai, India 01/18-10/22
Teaching Assistant
- High Performance Computing.
- Computational Fluid Dynamics and Heat Transfer.
- Heat Transfer Laboratory.
- Fluid Mechanics Laboratory.

Teaching interests: Fluid Mechanics; Heat Transfer; Computational Fluid Dynamics; Numerical Methods; Multiphase Flow; Fluid-Structure Interaction; Finite Element Methods; High-Performance Scientific Computing; Scientific Machine Learning and PINNs.

ADDITIONAL MACHINE LEARNING EXPERIENCE

Machine Learning Internship 03/22-05/22
Multi-agent trajectory prediction for autonomous driving
- Implemented the AgentFormer architecture in Python and generated semantic road maps from lane, drivable-area, and divider data for multi-agent trajectory prediction.

AWARDS AND HONORS

- **Naik and Rastogi Award** - IIT Bombay award recognizing excellence in Ph.D. research, 2024.
- **Best Ph.D. Annual Progress Seminar Award** - Department of Mechanical Engineering, IIT Bombay, 2022.

- **Dantec Prize** - 17th UK Heat Transfer Conference, 2021.
- **JICA Fellowship for Ph.D. studies at the University of Tokyo** - Awarded; declined.

PROFESSIONAL SERVICE

- **Peer reviewer:** Journal of Computational Physics; International Journal of Heat and Mass Transfer; Physics of Fluids; Journal of the Indian Institute of Science.

SELECTED CONFERENCE CONTRIBUTION

1. Nguyen, T.-B.; Liu, D.; **Raut, H. S.**; Bhattacharya, A.; Sharma, A.; Tran, T. **Enhancement of convective heat transfer using magnetically flapping fin array.** 17th UK Heat Transfer Conference (UKHTC 2021), Manchester, UK. Received the Dantec Prize.

COMPUTATIONAL AND TECHNICAL EXPERTISE

- **Numerical methods:** Finite-difference, finite-volume, finite-element, projection methods, Krylov solvers, immersed-boundary methods, level-set and VOF interface capturing, CFD-DEM, reduced-order modeling, stability analysis.
- **Programming and HPC:** C/C++, Python, MATLAB; MPI, OpenMP, CUDA, OpenCL; parallel CPU/GPU solver development; PETSc; large-scale scientific post-processing.
- **Scientific computing software:** OpenFOAM, CFDEM, LIGGGHTS, FEniCS/FEniCSx, PyTorch, TensorFlow, NumPy, SciPy, Pandas, scikit-learn, ParaView, Tecplot.
- **Research domains:** Unsteady CFD, FSI, multiphase flow, boiling and phase change, propulsion, particle-laden flow, porous-media transport, computational biofluids, DRL, PINNs, data-driven modeling.

PROFESSIONAL REFERENCES

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